

Centrality

Day 2

How do we determine who is
“important” in a Network?

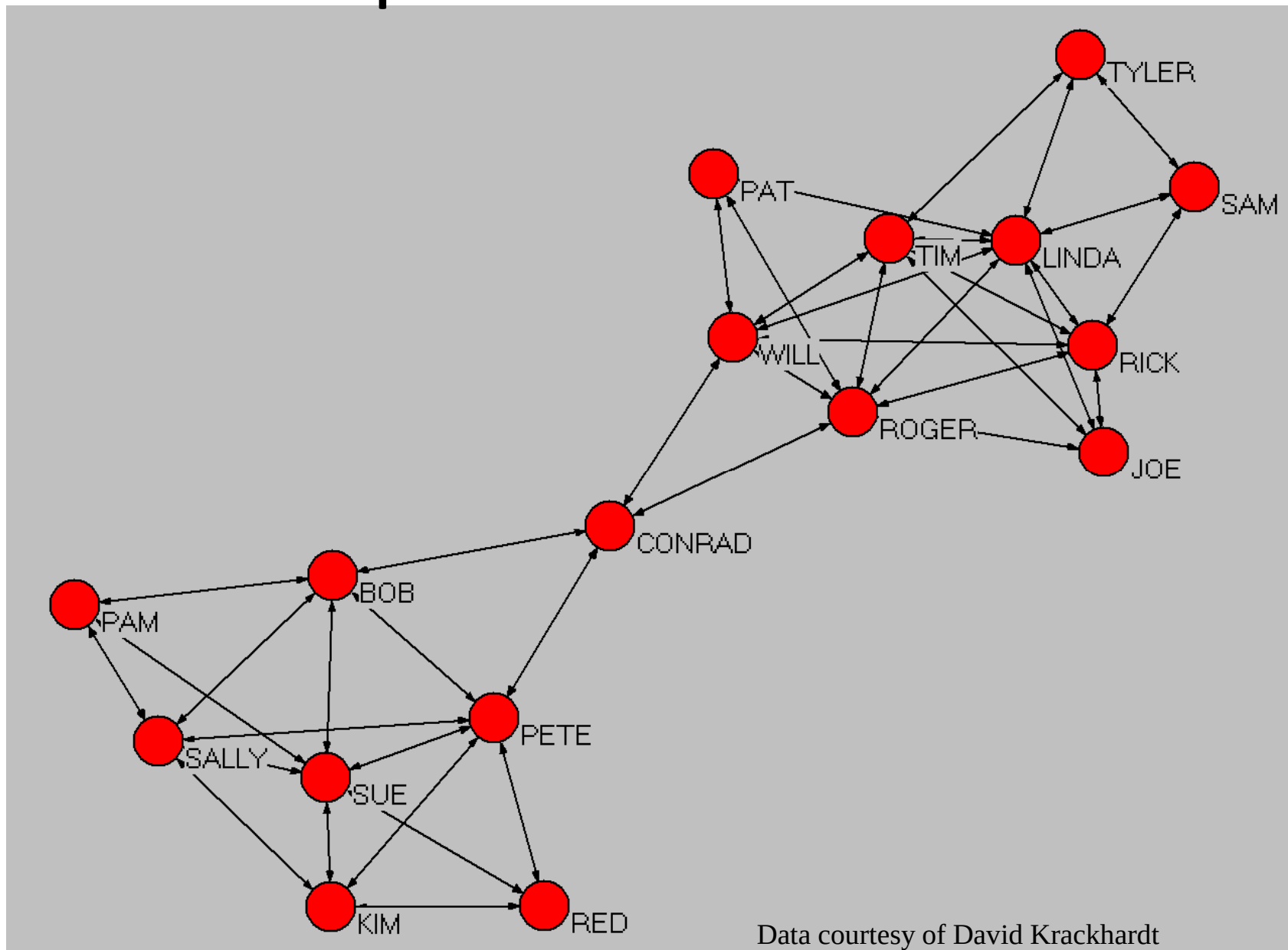
Centrality

- A measure of how network structure and position contributes to a node's importance
- Value associated with every node
- Many different measures which capture different aspects
- Can be characterized by the nature of the flow

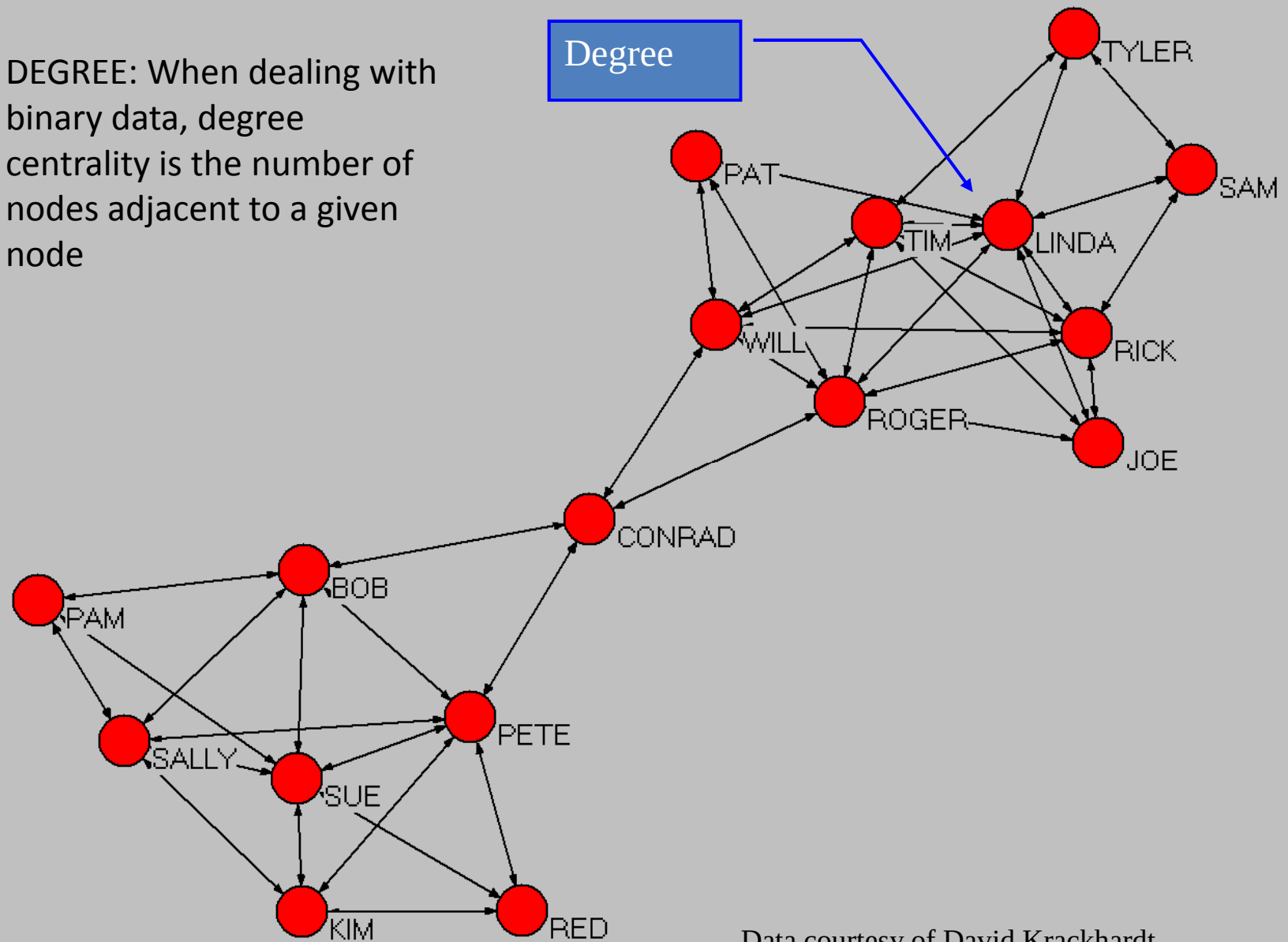
Centrality: different measures

- Degree
 - how well connected; direct influence
- Closeness
 - how far from all others
 - how long information takes to arrive
- Betweenness
 - brokerage, gatekeeping, control of info
- Eigenvector
 - being connected to the well connected (a popularity & power measure)

Who's Important in this network?



DEGREE: When dealing with binary data, degree centrality is the number of nodes adjacent to a given node



Data courtesy of David Krackhardt

Degree Centrality

- Index of exposure to what is flowing through the network
- Interpreted as opportunity to influence & be influenced directly
- Predicts variety of outcomes from virus resistance to power & leadership to job satisfaction to knowledge

Degree

	Bob	Con	Joe	Kim	Lin	Pam	Pat	Pete	Red	Rick	Rgr	Sally	Sam	Sue	Tim	Tylr	Will	
Bob	0	1	0	0	0	1	0	1	0	0	0	1	0	1	0	0	0	5
Con	1	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	1	4
Joe	0	0	0	0	1	0	0	0	0	1	1	0	0	0	1	0	0	4
Kim	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	0	4
Lin	0	0	1	0	0	0	1	0	0	1	1	0	1	0	1	1	1	8
Pam	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	3
Pat	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	3
Pete	1	1	0	1	0	0	0	0	1	0	0	1	0	1	0	0	0	6
Red	0	0	0	1	0	0	0	1	0	0	0	0	0	1	0	0	0	3
Rick	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	1	5
Rgr	0	1	1	0	1	0	1	0	0	1	0	0	0	0	1	0	1	7
Sally	1	0	0	1	0	1	0	1	0	0	0	0	0	1	0	0	0	5
Sam	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	1	0	3
Sue	1	0	0	1	0	1	0	1	1	0	0	1	0	0	0	0	0	6
Tim	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0	1	1	5
Tylr	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	0	0	3
Will	0	1	0	0	1	0	1	0	0	1	1	0	0	0	1	0	0	6
	5	4	4	4	8	3	3	6	3	5	7	5	3	6	5	3	6	

Degree Centrality with Directed Data

- Indegree- The number of ties directed to the node
- Outdegree- The number of ties that the node directs to others

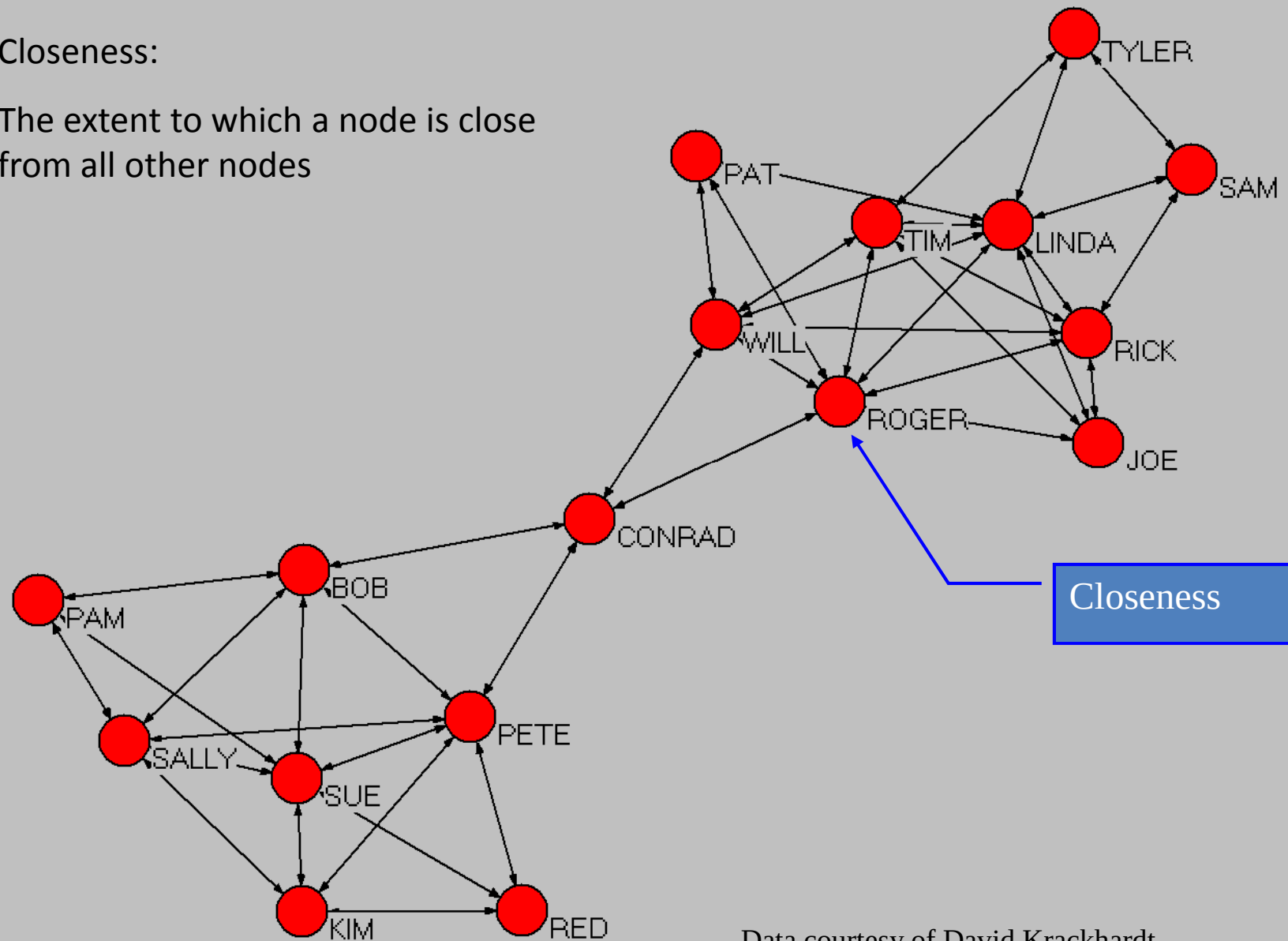
Degree Centrality with Valued Data

									OUTDEGREE
	MT6	MT71	MT72	MT83	MT93	MT210	MT215	MT272	
MT6	0	100	500	1600	1100	300	2450	1500	7550
MT71	0	0	0	0	0	0	0	0	0
MT72	0	0	0	0	0	0	0	0	0
MT83	0	0	0	0	0	0	0	0	0
MT93	0	0	0	0	0	0	0	0	0
MT210	0	0	0	0	0	0	0	0	0
MT215	0	0	0	0	0	0	0	0	0
MT272	0	0	0	0	0	0	0	0	0
INDEGREE	0	100	500	1600	1100	300	2450	1500	0

NOTE: NetDraw “binarizes” valued data when measuring centrality, UCINET does not.

Closeness:

The extent to which a node is close from all other nodes



Data courtesy of David Krackhardt

Closeness Centrality

- Is an inverse measure of centrality
- The extent to which a node is close from all other nodes
- Index of expected time until arrival for a given node of whatever is flowing through the network
 - Gossip network: central player hears things first, on average

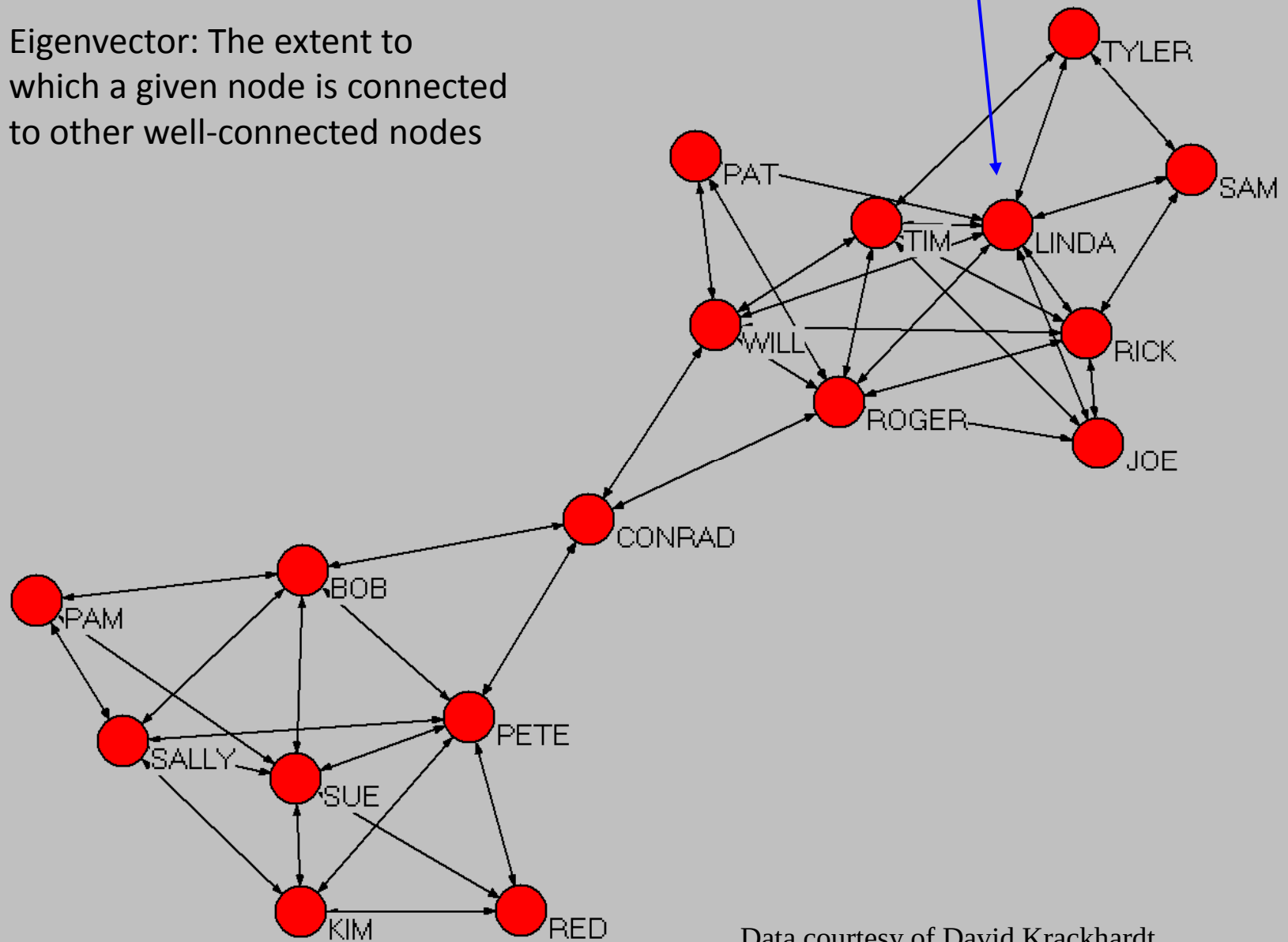
GEODESIC DISTANCE MATRIX used to Calculate Closeness

	Bob	Con	Joe	Kim	Lin	Pam	Pat	Pete	Red	Rick	Rgr	Sally	Sam	Sue	Tim	Tylr	Will	
Bob	0	1	3	2	3	1	3	1	2	3	2	1	4	1	3	4	2	36
Con	1	0	2	2	2	2	2	1	2	2	1	2	3	2	2	3	1	30
Joe	3	2	0	4	1	4	2	3	4	1	1	4	2	4	1	2	2	40
Kim	2	2	4	0	4	2	4	1	1	4	3	1	5	1	4	5	3	46
Lin	3	2	1	4	0	4	1	3	4	1	1	4	1	4	1	1	1	36
Pam	1	2	4	2	4	0	4	2	2	4	3	1	5	1	4	5	3	47
Pat	3	2	2	4	1	4	0	3	4	2	1	4	2	4	2	2	1	41
Pete	1	1	3	1	3	2	3	0	1	3	2	1	4	1	3	4	2	35
Red	2	2	4	1	4	2	4	1	0	4	3	2	5	1	4	5	3	47
Rick	3	2	1	4	1	4	2	3	4	0	1	4	1	4	2	2	1	39
Rgr	2	1	1	3	1	3	1	2	3	1	0	3	2	3	1	2	1	30
Sally	1	2	4	1	4	1	4	1	2	4	3	0	5	1	4	5	3	45
Sam	4	3	2	5	1	5	2	4	5	1	2	5	0	5	2	1	2	49
Sue	1	2	4	1	4	1	4	1	1	4	3	1	5	0	4	5	3	44
Tim	3	2	1	4	1	4	2	3	4	2	1	4	2	4	0	1	1	39
Tylr	4	3	2	5	1	5	2	4	5	2	2	5	1	5	1	0	2	49
Will	2	1	2	3	1	3	1	2	3	1	1	3	2	3	1	2	0	31

NOTE: if data is directed, you can calculate in-closeness and out-closeness centrality

Eigenvector

Eigenvector: The extent to which a given node is connected to other well-connected nodes

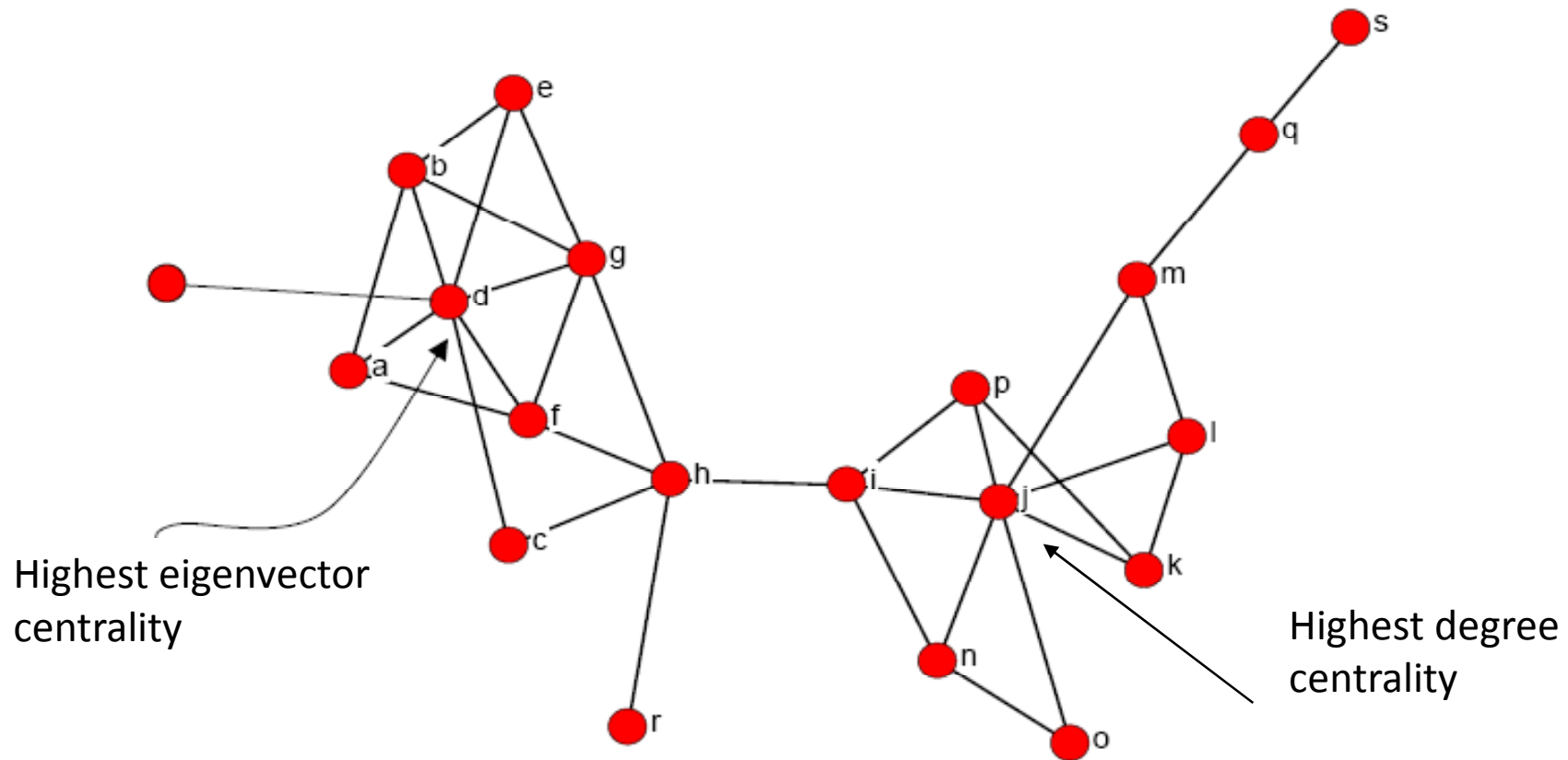


Data courtesy of David Krackhardt

Eigenvector Centrality

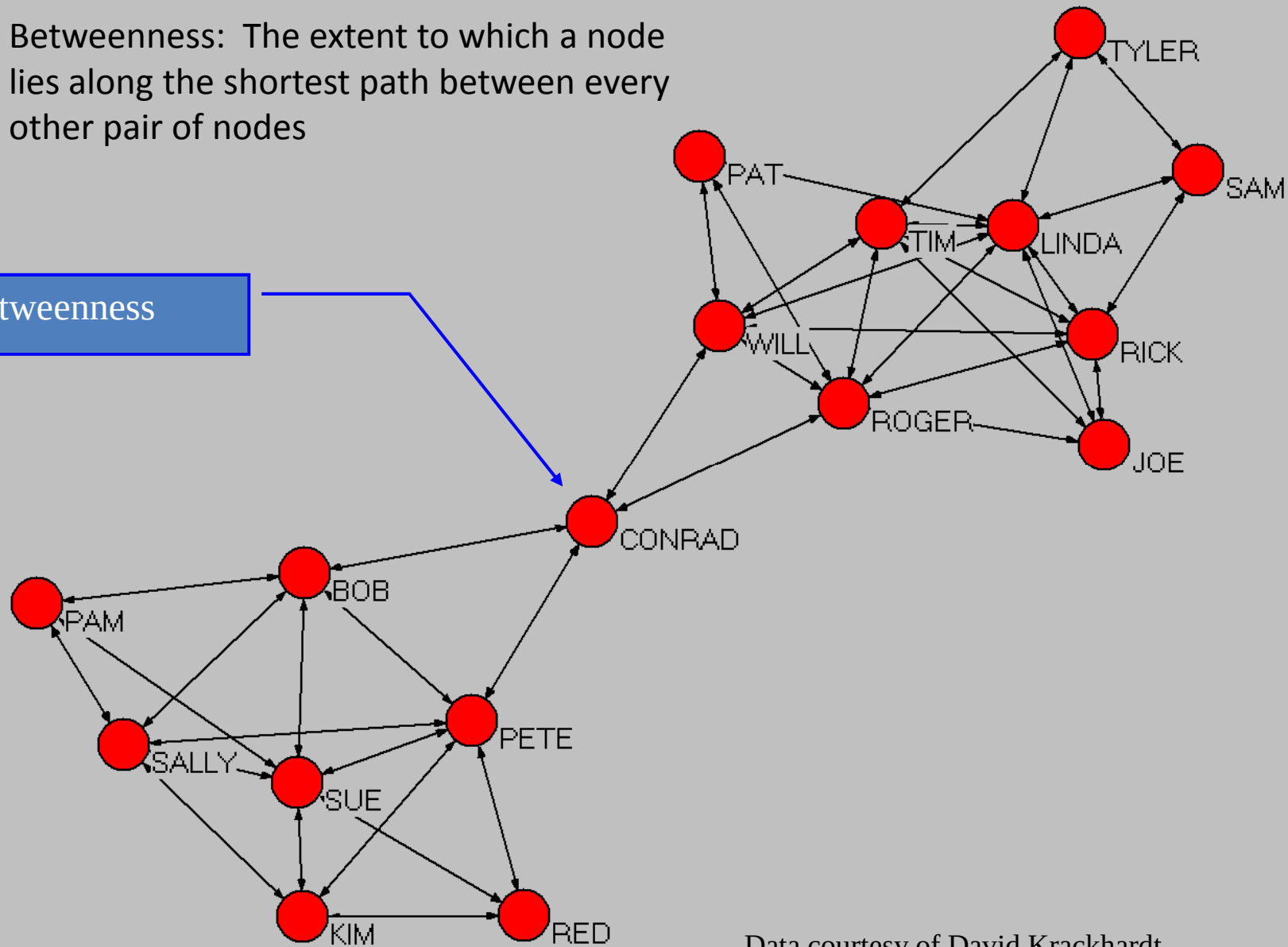
- Node has high score if connected to many nodes that are themselves well connected
- Indicator of popularity,
 - Google Page Rank
- Like degree, is index of exposure, risk
- However, tends to identify centers of large cliques

NOTE: Node with highest eigenvector centrality is not always node with highest degree centrality



Betweenness: The extent to which a node lies along the shortest path between every other pair of nodes

Betweenness



Data courtesy of David Krackhardt

Betweenness Centrality

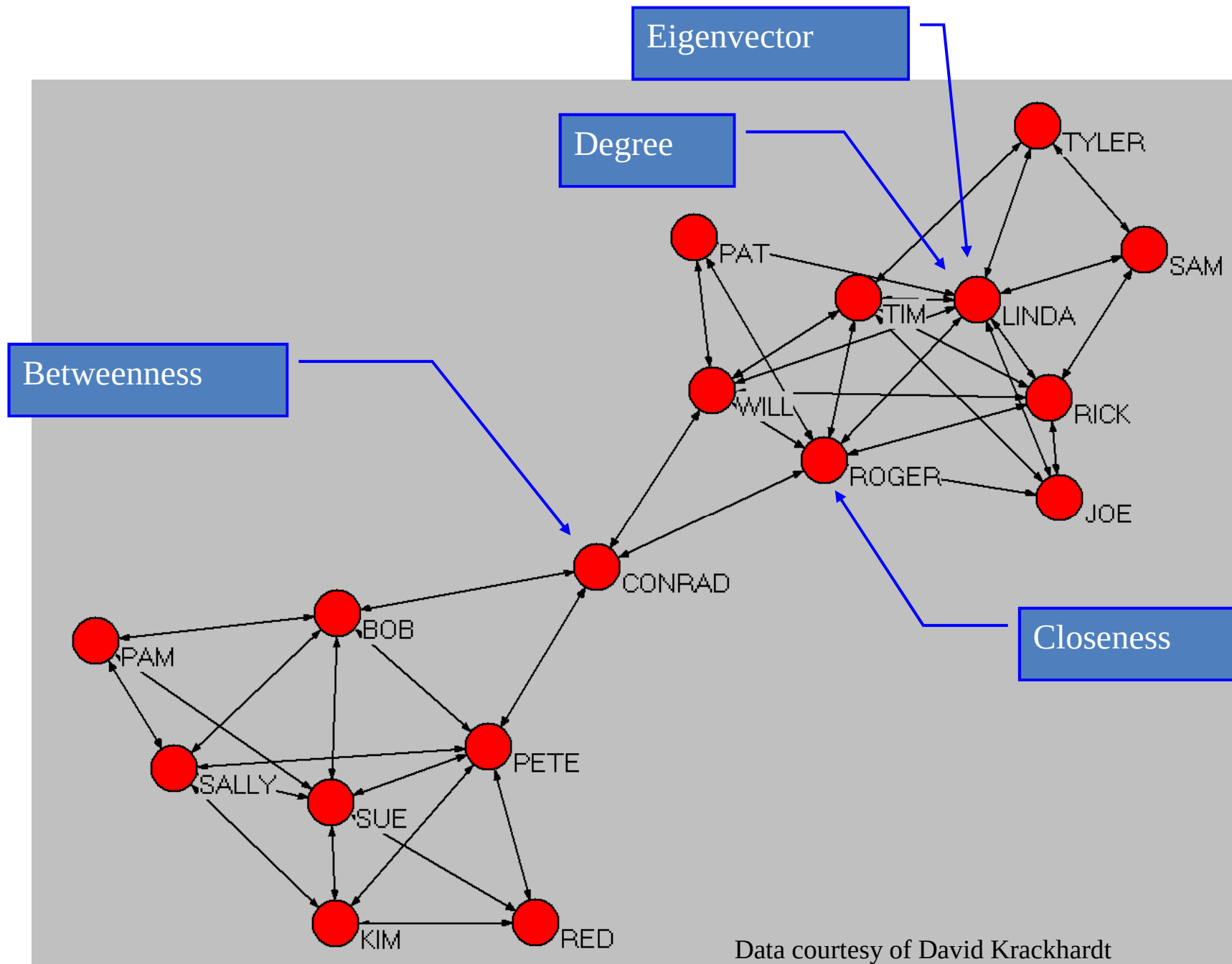
- How often a node lies along the shortest path between two other nodes

- Computed as:

$$b_k = \sum_{i,j} \frac{g_{ikj}}{g_{ij}}$$

where g_{ij} is number of geodesic paths from i to j and g_{ikj} is number of those paths that pass through k

- Index of potential for gatekeeping, brokering, controlling the flow, and also of liaising otherwise separate parts of the network
- Interpreted as indicating power and access to diversity of what flows; potential for synthesizing



Data Types: Centrality

	Disconnected or Connected	Binary or Valued	Directed or Undirected
Degree	Both	Both	Both
Closeness	Strongly Connected	Binary	Both
Betweenness	Both	Binary	Both
Eigenvector	Connected	Both	Undirected

Investigating Centrality

- With UCINET
 - Under Network | Centrality
 - Degree
 - Closeness
 - Betweenness
 - Eigenvector
- With NetDraw
 - Under Analyze | Centrality

Fun Applications of Centrality

- Oracleofkevinbacon.org
 - 6 degrees of Kevin Bacon
 - Can you find anyone with a Bacon score > 4 ?
- Theyrule.net
 - Board overlaps of top corporations
- Oilmoney.priceofoil.org
 - Tracking petroleum industry campaign contributions